

**PROJECT REPORT ON**

**BIOSCAN AI: A UNIFIED HYBRID VISION FRAMEWORK FOR  
HUMAN BODY WEIGHT ESTIMATION USING THE NEXT  
GENERATION VISION ARCHITECTURES**

*Report submitted to the SASTRA Deemed to be University as  
the requirement for the course*

**IMAGE PROCESSING & ANALYSIS**

**CSE608**

**Submitted by**

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### BONAFIDE CERTIFICATE

This is to certify that the report titled “**Bioscan AI: A unified hybrid vision framework for human body weight estimation using the next generation vision architectures**” submitted as a requirement for the course, **CSE608: Image Processing & Analysis** for M.Tech. Artificial Intelligence and Data Science programme, is a bona fide record of the work done by **Mr. Metha Shankar G A (Reg. No.127162016)** during the academic year 2025-26, in the School of Computing, under my supervision.

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## ABSTRACT

Nowadays, estimation of a human body weight from the image data is a challenging problem in computer vision and biometric analysis. Traditional methods of weight estimation/computation heavily rely on manual measurements or specialized hardware devices like weight machine which was found in hospital, gyms, and any public places. Recent advancements in technology and computer vision techniques have significantly improved the ability of machines to extract the meaningful biometrical features from the images. The integration of such next-generation vision architecture models allows complex human body features to be analysed freely. All the available existing systems are mostly focused on medical applications like for patients, but it cannot be useful for common man. Other systems which focused on visual image systems used simple regression models or ensemble models, however it doesn't use next-gen vision architectures. We used Celeb-FBI dataset which contains 7222 images which is sampled to 2500 images out of them 2492 images were selected after preprocessing and split into train, test and val for training purpose. This dataset is released in 2024 it is an under-explored dataset for weight estimation purpose so it is being used in our work. The proposed methodology consists of next-gen vision architecture models like Real time Detection transformer (RT-DETRV2) for detecting the person in the image with confident score, Segment Anything Model 3(SAM 3) model by Meta used to remove background noise in the image segmented by RT-DETRV2 and helps to extract the body area feature by using the on pixels in the image from the generated image by itself, You only look once (YOLO26-Pose) model used to compute the shoulder width, person height, torso length and hip width then these features are fed into the hybrid vision transformer model named FastViT by Apple to estimate the weight of the given person in the image. We used FastViT training for 50 epochs with early stopping patience 10; it stopped at 45 and achieved performance metrics Root Mean Squared Error (RMSE) of 8.36, Mean Absolute Error (MAE) of 6.37, Co-efficient of determination ( $R^2$  score) of 0.69, and Mean Absolute Percentage Error (MAPE) of 9.61%. These results show the significance of feature extraction and dataset in giving the best performance for our system.

**Keywords:** Human weight estimation, Computer Vision, Deep Learning, Vision Transformers, Image segmentation, SAM 3, RT-DETRV2, YOLO26-Pose, FastViT.

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### List of Abbreviations

| Abbreviation | Full Form                       |
|--------------|---------------------------------|
| MAPE         | Mean Absolute Percentage Error  |
| MAE          | Mean Absolute Error             |
| MSE          | Mean Squared Error              |
| RMSE         | Root Mean Square Error          |
| RT-DETRV2    | Real-time detection transformer |
| YOLO         | You Look Only Once              |
| SAM 3        | Segment Anything Model          |
| FastViT      | Fast Vision Transformer         |
| BMI          | Body Mass Index                 |
| AI           | Artificial Intelligence         |
| CNN          | Convolutional Neural Networks   |
| GPU          | Graphical Processing Unit       |
| AMP          | Automated Mixed Precision       |
| KDE          | Kernel Density Estimation       |

# 1 Introduction

Human body weight is a crucial component playing vital role as well as act as indicator of the person's fitness level and overall health and their physical condition. Most of the disease in the world, about almost 90% of them caused by the overweight and obesity people (Jan et al., 2025). Therefore, tracking the human body weight then and there constantly reminds the individual about the weight so that he will be more conscious about them and helps him to do something with it. In recent days, many people aware of the consequences of overweight and obesity started doing all kinds of workouts, diets, and stuff like that. They want to track their progress, which is measuring their weight weekly to know whether the workout plan and the diet plan work for them or not. It also adds a lot of motivation to pursue the progress that they have taken (Kim et al., 2023). Medical professionals need to know the person weight as it is important to conduct several tests followed by that.

Traditional methods of weight estimation/computation heavily rely on manual measurements or specialized hardware devices like weight machine which was found in hospital, gyms, and any public places (Velardo & Dugelay, 2010). For common person it is highly complex to know the weight like we can't measure every single daily if we need to measure daily then buying a weight machine is the option which too costs. Therefore, if we have developed the system where from just an image a layman can able to see his weight accurately every now and then he wants. Thus, the development of an image-based biometric framework using Next-Gen Vision Architectures for weight estimation has become a critical research direction.

Nowadays, estimation of a human body weight from the image data is a challenging problem in computer vision and biometric analysis (Wen & Guo, 2013). Because, the image will have noise in it, so removal of those will take time and worth doing it. The images that we have taken is not restricted to medical use alone since it doesn't have much noise in the image but for the common man, they may take photo of them in home, college, or anywhere they want even it includes the crowd. The problem that we have is to estimate the weight of the one-person while removing others in the background. Also, eliminating the background noise like poster, trees, walls, etc. don't be meaningful to have so removal of them also important.

All the available existing systems are mostly focused on medical applications like for patients, but it cannot be useful for common man. Other systems which focused on visual image systems used simple regression models or ensemble models, however it doesn't use next-gen vision architectures (Tirink et al., 2023). Moreover, most of the applications out there used for medical purposes alone not for common people frankly Celeb-FBI dataset is not explored yet to estimate human body weight from 2024 when it released to till the point (Debnath et al., 2024). Most of the applications, uses deep learning architectures then statistical models and the hybrid vision transformer for weight estimation is not much explored. Not many efforts dedicated to feature extraction as it is the important part of the project.

To develop the system which correctly removes the background noise, segmenting a person and estimating their weight may look simple, but a lot of techniques needs to be employed in order to achieve that, but it can't be achievable just by using deep learning models, statistical models or ensemble models we need much more sophisticated models which performs better gives high accuracy using less computational resources (Ashmawi et al., 2019). The system where background noise removed, segmenting the person correctly in the image, pose estimation of the person in the image using pose detection models finally estimating the weight from the cleaned image and the extracted features from the models.

The human weight estimation from the images poses several challenges due to variations in the real-world environment. There are several factors which affects the estimation of the human body weight from images. Factors such as clothing style, this can significantly affect the performance of the model since it perceives the body shape in a different manner like as loose or layered clothing may have the possibility of hiding the person's underlying human body structure. Likewise, lighting conditions this is another important factor to be addressed earlier itself as poor lighting or shadows may have the chance of distorting the visual features of the given image.



Camera angles can also affect the performance of the model, since images captured from the different viewpoints may or may not represent the true structure of the human body. In addition to that, another one of the most challenging factors in affecting the estimation of human body weight is the background complexity and the presence of multiple objects or people can make it very difficult for the system to isolate the person from the complex environment. The removal of crowd and the multiple objects present in the image has to be removed as it is not adding any value to estimate the weight. Since crowd consists of various people with varying lighting conditions, using different postures, using different clothing styles and most importantly the system finds very difficult to estimate the weight in this scenario and posing serious performance issues by not improving the performance of the system (Xiaohui et al., 2018).

Accurate weight estimation from the images heavily relies on the quality of the features extracted from the given data. The simple statistical models and the deep learning models may not be sufficient to capture the complex relationship between shape of the body and the weight. Hence, it is crucial for extracting meaningful features such as area of the body, shoulder width, hip width, torso length becomes essential. A robust feature extraction pipeline can simplify the process as well as the performance of the system, as it helps in improving the reliability and accuracy of the predicted weight (Chicco et al., 2021). Recent advancements in technology and computer vision have significantly improved the ability of machines to extract the meaningful biometrical features from the images. Technologies like object detection, pose estimation, segmentation and classification models have enabled us to accurately extract the relevant features from the human body image. These techniques are widely used in various industries like medical diagnosis, transport like traffic detection, police security systems like surveillance system, human body measurements in fashion industry. The integration of such models allows complex human body features to be analysed freely (He et al., 2017; Kirillov et al., 2023; Cao et al., 2017).

The proposed framework solves the problem by adopting a unified hybrid vision framework for human body weight estimation. It uses the images taken from Kaggle Celeb-FBI dataset, which consists of celebrity images with all the possible noises found in the background as well as the crowd. The flow of our proposed framework goes from segmenting the person in the image, extracting the person with the high confidence score in the image while removing the background noise by giving black colour to the background, pose estimation using the pose detection models to avoid the problems like Zoom-in problem and finally using the hybrid vision transformer to estimate the weight from the extracted features and the cleaned images (Lv et al., 2024; Carion et al., 2025; Maji et al., 2022; Dosovitskiy et al., 2020). The strength of our proposed framework is that it uses the next-generation vision architectures which are released in the past 5 years. It is highly optimized, improved, developed by the key industry players like Baidu, Meta, and Apple which makes the models more valuable to our framework thus creating the better outcome. The industry used models are promising to give the high-performance with low latency, and we used them wisely in the system. We have employed a highly extensive approach wholly dedicated to the feature extraction itself. Feature Extraction remains crucial in the weight estimation problem, and we have solved them by using various techniques available for segmenting, detecting and extracting the relevant features from the given data (Rokhim et al., 2025).

The implementation of a system that can able to estimate the human body weight from the digital images has revolutionary benefits for a variety of industries, ranging from the large-scale medical diagnosis to the customized fitness app. Beyond the simple elimination of the physical hardware weighing machines, such a system addresses the major impact of making the healthcare services available to anyone who has the smartphone devices. It shifts the idea about the smartphone from entertainment to sophisticated biometric system. The people living in rural or resource constrained environments where physical weighing machine scales are often inaccessible or rare maintenance of such machines as it requires frequent checks and services. This system provides the digital image-based approach provides a low-cost, scalable alternative for the basic health assessment. The integration of such systems even further allows for continuous health monitoring, thus enables clinicians to track the patient progress remotely. This is especially helpful for elderly patients or individuals with disability or limited mobility who find it difficult in travelling or even standing. The fitness and sports industry this real-time human body weight estimation can significantly enable the players or athletes to know their physical metrics which helps to improve their current workout plans and the diet plans instantly (Manichand et al., 2026).

## 2 Related Works

In the age of artificial intelligence (AI), there are numerous significant research efforts have been directed towards estimating human body weight using various techniques. With the rapid growth of technologies like machine learning, deep learning techniques, and the transformer models. There are several approaches have been proposed to analyse the digital image data and extracting meaningful information related to the human body structure. Researchers have been explored a wide range of methods, from statistical models like linear regression, Random Forest Regressor, Support Vector Regressor, etc. to advanced deep learning architectures like ResNet, VGG16, etc. to improve the accuracy and reliability of such estimations. These studies have contributed to the development of various techniques for feature extraction, image processing and pattern recognition. Despite the progress made so far, there are still several challenges in achieving accurate and robust performance in the real-world environment. This chapter deals with reviewing the existing literature and highlighting the key approaches used in it.

Manichand, F. R. et al., (Manichand, F. R., 2026) proposed a method to estimate body mass index (BMI) from image using WayBED dataset. It uses feature extraction techniques like 17 skeletal key point detection, Principal Component Analysis (PCA), and K-Means Clustering to filter the data by removing irrelevant images. The challenge is that dataset is imbalanced and weak feature extraction. Sapkota, R. et al. (Sapkota, R., 2025) this paper presents an overview of the evolution of the family of object detectors such as YOLO models. It highlights key innovations made in the architectures for to stable the model training. It benchmarks its models based on the COCO dataset which provides a detailed comparison among the models such as YOLOV5 to YOLO26 models. The performance metrics needed to evaluate the models are all used along with the inference speed to showcase the trade-offs between accuracy and efficiency. It also discusses about the real-world applications of these models in fields like robotics, agriculture, etc.

Shams, M. Y. et al. (Shams, M. Y., 2025) this paper specifically estimating the weight of the broiler using the external appearance instead of body size. The YOLOV8 model has been employed here in this paper. Various data augmentation techniques used for the feature extraction. The challenges in this paper are it does not use pose estimation which could be helpful in key point estimation thus improving the performance of the model. Finally, it does not use multiscale feature extraction just simple data augmentation techniques has been used instead of advanced feature extraction. Jan, M. T. et al. (Jan, M. T., 2025) this paper presented a comprehensive survey and comparison of body weight estimation techniques, datasets and its applications. The paper highlights the different works done using deep learning and statistical models by using various datasets. Even though the performance of their work is well, but it suffers deployment issues since it takes more computational resources. The Celeb-FBI dataset is not used in this paper for comparison. It does not use the latest models which have the potential of deployment and hybrid vision transformer has not utilized in any of the works presented in this paper.

Carion, N. et al., (Carion, N., 2025) this study introduced the SAM 3 model that can able to detect, segment and also tracks the objects in both the images as well as in the video. It focuses primarily on how the several changes made in the architectural level, ease of deployment and the inference speed which boosts the accuracy of detection and improving the capabilities of SAM model on segmentation tasks. Bharadwaj, S. et al., (Bharadwaj, S., 2025) this paper estimates the body weight of the pig using the digital images. It uses SAM for extracting the features for precise segmentation also various deep learning models compared against the various statistical models. After the comparison, Xception model found to be the best performer model out of all others. The challenge in this paper is it is limited to animals not humans also does not focus on pose estimation and the use of hybrid vision transformer models to estimate the weight.

Rokhim, I. F. et al. (Rokhim, I. F., 2025) this paper detects the fire and smoke using the benchmarked dataset. The YOLO models and the transformer-based RT-DETR models have used and compared. The results shows that the RT-DETR performs better in real-time detection tasks than YOLO models. However, it is not applied to human body weight estimation and biometric analysis tasks. Debnath, P. et al., (Debnath, P., 2024) this study addresses the need for the full human body datasets with labelled height, weight, age, and gender. Since the dataset is

significant in industries like sports, Virtual Reality (VR), fashion and design industries. The dataset is prepared in a way such that it will be a balanced one. Using the dataset they compared three deep learning models: ResNet-50, VGG16 and Convolutional Neural Network (CNN) among them ResNet-50 outperformed all others. The challenges are that not much feature extraction techniques have been utilized in this study.

Lv, W. et al. (Lv, W., 2024) this study presents the upgraded version of RT-DETR model which is RT-DETRV2 model. It is designed in such a way that it can be easy to deploy by removing deploying constraints associated with DETR model, thus optimizes the training strategy. It enhances the performance of the model since some of the changes made specific to RT-DETR models when compared to other YOLO models. Zhu, Z. et al. (Zhu, Z., 2024) this paper proposed SAM model for segmentation by adapting this powerful model extends to not only segmentation but also pose estimation. The proposed method shows that SAM can easily understand the structure of the human body, and it performs better in both segmentation and pose estimation. However, the SAM model have computational complexity in feature extraction, which may limit its potential in deploying the applications in the real-world environment.

Tırink, C. et al., (Tırink, C., 2023) this paper proposed a method where it is aimed to estimate the body weight of the herd in Poland. Numerous features have been used in this work also it shows the important features needed for estimating the body weight of the herd in Poland. They used Classification and Regression trees algorithm (CART), Random Forest regression and support vector regression. However, the challenges in this work is that they have used only simple regression models, not the advanced deep learning models and the transformer-based models. Kim, J. et al. (Kim, J., 2023) this paper utilizes wearable sensor data which is obtained from smart insoles. The approach focuses on multitask, including the activity recognition speed estimation and weight estimation using MobileNetV2 model as backbone network. However, this work restricted to only wearable sensor data and not used for non-contactable sensors like digital image data.

Vasu, P. K. A. et al. (Vasu, P. K. A., 2023) this study proposed a model named FastViT, a hybrid vision transformer model which integrates the strengths of transformer and also the designs of the convolutional networks. The RepMixer is the building block of the FastViT architecture. It increases the accuracy and also efficiency of the model. This study compared this model with all the other available models, but it outperforms and stands best among them. It can be used for classification, segmentation, and detection. Due to its extensive feature extraction capability, it can be even further extended to do regression tasks as well. Kirillov, A. et al. (Kirillov, A., 2023) this study presents a new segmentation model named SAM. They have built the large dataset for the segmentation purpose when compared to the all the other available datasets found so far. They conducted several experiments using the dataset and the SAM model, and they impressed with the zero-shot performance of the model for the given dataset.

Maji, D. et al. (Maji, D., 2022) this study introduced a new pose estimation model named YOLO-pose, a novel heatmap-free approach used to estimate the pose of the person found in the image using the bounding boxes. It performs better than the previous state-of-the-art techniques to estimate the pose of a person in the digital image. It is validated using the COCO dataset; therefore, it can be used for real-world applications. Chicco, D. et al. (Chicco, D., 2021) this study discusses about the performance metrics that can be used for regression. For regression tasks no one metric can tell the performance of the regression itself unlike classification tasks. They explored various performance metrics available for regression like Mean absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE),  $R^2$  score, Mean Absolute percentage error (MAPE) and symmetric Mean Absolute percentage error (SMAPE). Among them,  $R^2$  score found to be as standard performance metric to evaluate the regression model.

Dosovitskiy, A. et al. (Dosovitskiy, A., 2020) this study discusses about the efficiency of the transformer architecture over the CNN architecture. Many previous works are using CNN architectures alone or else as ensemble model. In this study, CNN is not necessary by applying a pure transformer model applied directly to the dataset and performs very well on the image classification tasks. It also performs very ell than the previous state-of-the art models. Ashmawi, S. et al., (Ashmawi, S., 2019) this paper proposed a method to estimate the human

body measurements from real-time human digital images using the Haar Cascade classifier and Support Vector Machines (SVM). It is mainly focused on solving the online shopping platforms problems for clothing. However, the work just used machine learning models and the dataset contains very controlled environment images.

Feng, M. et al. (Feng, M.,2019) this paper developed a system where it estimates the weight of the foetal using low birth weight (LBW) and high birth weight (HBW). Initially, the dataset is imbalanced then applied synthetic minority over-sampling techniques (SMOTE) to make it balanced. SVM is used for classifying the foetal, then estimated the weight of the fetal using Deep Belief Network (DBN) model. The challenge here is this system is restricted to only fetal weight estimation. Xiaohui, T. et al. (Xiaohui, T., 2018) this paper developed a system where it was able to estimate the human body size such as shoulder width, bust girth, hips, and waist using Random Forest regression. The feature extraction technique used here is Random Forest Regression with geodesic distances method. The main challenge here is the pose estimation models have not been explored.

He, K. et al. (He, K.,2017) this study presents a simple, flexible framework for segmentation model name Mask R-CNN. Mask R-CNN model extends the previous Faster R-CNN model by adding certain features in the architecture of the model. It shows better performance compared to other models. The main challenge is deployment seems difficult as it takes more computational resources. Cao, Z. et al. (Cao, Z., 2017) this study presents an approach to estimate the 2D pose of multiple persons found in the image using the specialized technique called Part Affinity Fields (PAF). PAF can able to perform well and gives better accuracy in the real-time performance regardless of the number of people in the digital image. The challenge here is it is not applied in scenarios like complex scenes.

Wen, L. et al., (Wen, L., 2013) this paper discussed about the correlation between the facial features and Body Mass Index (BMI). It develops a system where it uses MORPH-II dataset then applied basic feature extraction techniques. The BMI estimation task done by the SVR model. The main challenge here is the dataset is in controlled environment, simple feature extraction techniques applied and the simple regression model. Velardo, C. et al. (Velardo, C., 2010) this paper discussed about the importance of weight estimation as it is a crucial biometric trait which is widely used in medical and forensic domains. It initiated the idea of estimating the human body weight from the digital image. They have explored the large medical database using multiple regression models. The challenge here is simple regression models alone can't be sufficient. Jain, V. et al. (Jain, V.,2010) this study proposed a benchmark dataset for face detection as the previous methods lack common evaluation methods and hasn't captured some of the aspects of the image found in complex real-world scenarios.

### 3 Proposed Framework

The proposed framework estimates the human body weight using the Celeb-FBI dataset, which is released in 2024. The segmentation tasks initially done by the YOLO26-seg model which is released in 2026, but it found to less efficient and giving inappropriate results which could affect the performance of the system. After that, a high-precision transformer-based architecture developed by Baidu and released in 2024 named RT-DETRV2 has been employed here to the high-precision segmentation, which could impact the performance of the model. SAM 3 released in 2025 developed by Meta used to segment the person from the background noise. YOLO26-Pose model released in 2026 used to pose detection of the person in the cleaned SAM 3 image. The feature extraction has been done by using the cleaned SAM 3 images and YOLO26-pose model labels. The extracted features had been fed into FastViT model to estimate the weight of the human body in the image. The overall workflow of the proposed system is depicted in Figure 1, illustrating the end-to-end pipeline.

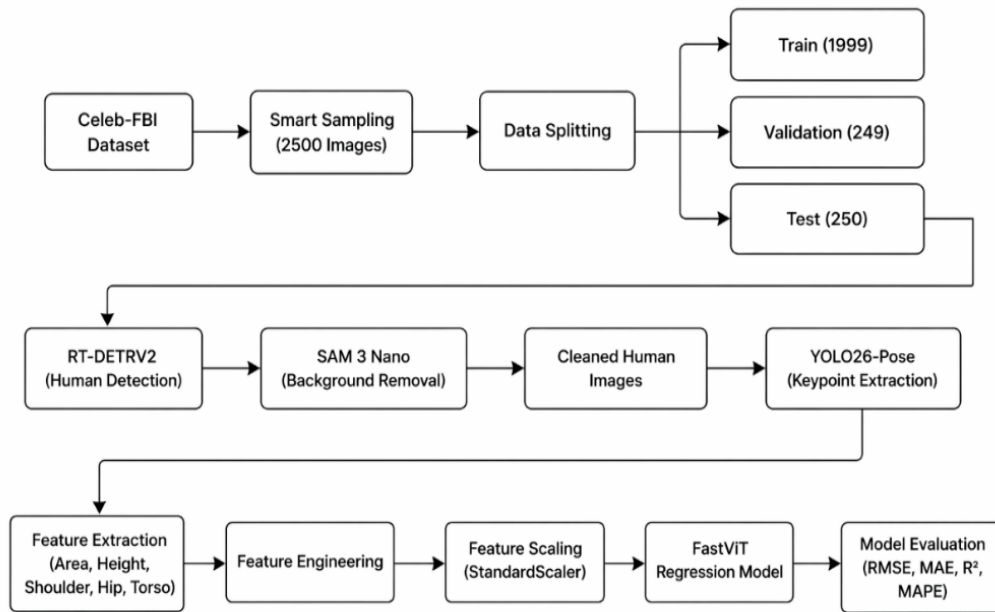


Figure 1: Overall workflow of the proposed human weight estimation framework

The YOLO26-seg model was initially used to segment the person in the image. It is not an object detection model, whereas it is an instance segmentation model which can segment the object with assigning the class labels and the confidence score for that class. It ensures the information about the precision of the segmentation. Instance segmentation model like YOLO26-seg helps in places where need to know not only the object found in the image but also to know the exact shape of the model. It promises to give high accuracy and the efficient model. Despite all these strengths, for our proposed work this instance segmentation doesn't seem to help. Because, some of the results were not as satisfied as it is expected. It succeeds some in some of the cases and identifies the person with the highest confident score with the proper bounding box surrounded around them. For our proposed work, we need high-precision instance segmentation model as this model doesn't meet our standards it has been removed from our pipeline but still it results are stored in the Google Drive. The results are stored in Google Drive under the folder named segmentation\_masks, and the labels are also stored in the separate folder generated by the model.

Since, we have planned to use only the next-generation vision architecture model the transformer-based object detection model named RT-DETRV2 developed by Baidu has been chosen for object detection. It is highly helpful and be useful in scenarios where needs to have high-precision object detection also at the same trying to perform really well in the real-time performance. It builds on top of the vision transformer architecture also it is an advanced real-time object detector. It was highly efficient in processing the multiscale features, thus helping in

detecting the object even as the complex background scenes may be present in the image. The query selection method employed in this model made this to focus on the most relevant object. It is a global-context awareness model so it finds easy to comprehend the object's meaning found in the image. The inference speed is high in real-time performance because of the dynamically adjusting decoding layers without the need for retraining. The RT-DETRV2 model large variant is being employed in this work. It has been given with the hyperparameters needed for the better performance. The confident score 0.45 given to ensure we get only the real humans, not the posters. The Intersection of Union value is set to be 0.45 for tighter boxes as it outputs are passed to the SAM 3 model. The results generated by the model, both the images and labels, were stored under the folder in the Google Drive. These outputs are going to be the inputs of the SAM 3 model.

The RT-DETRV2 model results are used in SAM 3 model to get the cleaned dataset. The SAM 3 model is the latest model, and it requires the authorization from the open-source platform HuggingFace. After the approval of the request from the huggingface, then it was integrated in ultralytics package from there we can able to use SAM 3 model effectively. The SAM 3 model is a promptable concept segmentation model where it takes the text as prompt or the exemplars of the image or combination of the both also can be used. It builds upon the previous SAM model which can segment only single objects per the given prompt, whereas the newer version SAM 3 model can able to find and segment every possible occurrence of a concept appearing in the image, which can be anywhere. SAM 3 achieves better performance gain than the existing PCS models. In our work, the prompt that we have given is the RT-DETRV2 images with the bounding box around the given objects, and it picks up the object with the highest confidence score while making the background of the image to black colour. The SAM 3 architecture consists of the core components detector, tracker, and the presence token, which shares a Perception Encoder (PR) vision backbone. The detector is based on the DETR-based architecture for the image-level concept detection and takes the given RT-DETRV2 image. The detector in the SAM 3 architecture takes this image, then outputs the mask head for the instance segmentation masks. The tracker in the SAM 3 architecture is a memory-based segmentation, which inherits its capability from the SAM 2 architecture. It acts as a memory bank for storing the object appearance found across the given frames. The final core component in the SAM 3 architecture is presence token which helps us to predict whether the human is present in the image or not which separates the recognition from localization thus improving the detection. Finally, we get the cleaned image which can be used for the further works in our work.

The resultant cleaned images generated by SAM 3 model will be fed into the latest pose estimation model named YOLO26-Pose. The pose estimation model has been employed in our work due to the zoom-in problems, clothing variations and poor lighting conditions. For example, if the person in the image is actually lean body type having 55 kg weight but wearing the loose hoodie. In this scenario, the previous models only able to detect the person with the highest confident score in the image also helps us to know the body area. The challenge here is due to that loose hoodie the 55 kg person will looks like 75 kg person to the model hence it will affect the performance of the system. Thus, we have employed YOLO26-Pose model, which can promisingly reduce the chances of this miscomputation errors. The YOLO26-Pose model identifies the location of specific points in an image, which are usually referred to as key points. The key points in the image represents various parts of the person, such as the joints. The representation of the key points will be like 2D or 3D co-ordinates. The 2D co-ordinates help us to know the joint position of the person in the image, whereas 3d co-ordinates helps us to know visibility as well along with the joint position. There are 17 key points each represents different parts of the human body structure. The YOLO26-Pose nano variant is used in our work to mainly extracting the key points from the image so that it will be useful features for our weight estimation system.

Feature Extraction is the most important part of our work since it will significantly affect the performance of the system. The SAM 3 cleaned images and the YOLO26-Pose model labels are used in our work to extract the features. There are five features extracted from them namely body area, height, shoulder width, hip width and torso length of the person. The body area extracted from the SAM 3 cleaned resultant images, it will compute the number of foreground pixels found in the cleaned image. The body area is useful since it conveys that more body are means generally more body mass. It will act as a proxy for body size. The height of the person is extracted from the YOLO26-pose labels, it will compute height of the person using topmost key point head to bottom key

point ankle pixel distance. It will also be helpful to normalize height and weight are the strongly correlated variables. The shoulder width of the person is extracted from the YOLO26-Pose labels, it will compute the shoulder width by left shoulder to right shoulder distance. The distance between 5th key point and 6th key point helps us to compute this. It is useful to know since it indicates the upper body structure. The broad shoulder usually conveys that the more body mass possibility. The hip width of the person is extracted from the pose labels; it will compute the distance between left hip to right hip. The distance between 11th key point and 12th key point enable us to compute the hip width. It is useful since it helps us to know the person's lower body mass in the image. It also reflects the fat distribution of the person in the image. The torso length is extracted from the pose labels; it will compute by shoulder width to hip width distance. The distance between 5th,6th key points and 11th,12th key points enable us to compute the torso length of the person. It is useful to know since it helps us to understand the body proposition of the person. These extracted features are combined into the previous CSV files created earlier and stored in the Google Drive separately for train, test and val.

Now, we move on to the core component of the weight estimation framework, the model building phase. Before, moving onto the model building phase, there is one important step to complete which is the feature engineering phase. Feature engineering simply means to convert the available data to meaningful, informative features. It takes up to 60-80% of the data professionals to improve the predictive power. The combined three CSV files for train, test and val are all are proceeded to the further processing. In our dataset, we found out that certain image has the weight value as zero. In real-world scenarios there is no possibility of a person having 0 kg weight, even a newborn baby weighs around 3.5 kg at birth. Since, it is irrelevant to have this data in our model, we have removed this from the CSV files. The three CSV files have the same structure having the common features namely filename, height, weight, gender, age, body\_area, p\_height, sh\_width, hip\_width and torso\_len. The model only accepts the numerical features if any categorical variables it needs to be converted into numerical ones when it is helpful to improve the performance of the model. However, the filename and gender are not so helpful and meaningful to improve the estimation of the human body weight so both of them have been removed from our CSV files. The numerical features are all in the different range of values, so the standard scaler normalization technique has been applied to normalize the values of the numerical features to improve the performance of the model. The feature engineering phase gets completed and ready to move on to the next phase.

The FastViT sa\_24 variant is used in our proposed work. The FastViT is a hybrid vision transformer model which efficiently combines the feature extraction capabilities of the CNNs and to capture the global contexts effectively using the transformer architecture. The main thing which separated this model from others is the introduction of the concept called structural reparameterization. The previous models have a lot of complex layers it has multiple paths, skip connections and take a lot of memory to store that in the training time but in the testing time it usually doesn't use much memory since it already learns in the training phase by combining multiple paths into the single one. In the FastViT model, in training time like previous models it takes multiple paths and skip connection to understand the underlying patterns but in the inference all the multiple paths, skip connections and the branches merged into single convolutional layer hence the structural reparameterization helps to reduce the access memory costs. The hyperparameters and the configurations employed in the FastViT model were given in the Table 1.

Table 1: Hyperparameters and architectural configurations

| Parameter                | Value / Configuration                          |
|--------------------------|------------------------------------------------|
| Model Architecture       | FastViT (fastvit sa24)                         |
| Pretrained Weights       | Yes                                            |
| Image Resize             | 224X224                                        |
| Batch Size               | 16                                             |
| No.of.epochs             | 50 (Early stopping used)                       |
| Optimizer                | AdamW                                          |
| Learning Rate            | $3 \times 10^{-4}$                             |
| Loss Function            | Mean Squared Error (MSE)                       |
| Learning Rate Scheduler  | ReduceLROnPlateau (patience = 5)               |
| Early Stopping Patience  | 10 epochs                                      |
| Dropout rate             | 0.3                                            |
| Image Feature Dimension  | 512                                            |
| Tabular Feature Layers   | 128 $\rightarrow$ 64                           |
| Fusion Strategy          | Concatenation                                  |
| Output Layer             | Fully connected regression layer with unit = 1 |
| Activation Function      | Rectified Linear Unit (ReLU)                   |
| Horizontal Flip          | Yes                                            |
| Brightness               | 0.2                                            |
| Contrast                 | 0.2                                            |
| Normalization            | ImageNet Mean & Std                            |
| Mixed Precision Training | Enabled (AMP)                                  |

The proposed framework presented an end-to-end pipeline with an efficient approach for estimating the human body weight using the Celeb-FBI dataset. The Celeb-FBI dataset is efficiently sampled and split to train, val and test. The proposed system uses the various image processing techniques including object detection, segmentation, pose estimation and the regression to achieve the better performance. In the beginning, the images are processed and segmented using the RT-DETRV2 model to detect the human with the confident score. The resultant image from the RT-DETRV2 model is then further refined by using SAM 3 model to isolate the human with the highest confident score, then removed the background noise. The cleaned SAM 3 images are then given to the YOLO26-Pose model, which gave 17 key points of the person in the image. Feature extracted from the cleaned images obtained from SAM 3 model and YOLO26-pose labels. The five features are extracted from them are height, hip width, shoulder width, torso length and body area. Feature engineering applied to the all the three CSV files. Finally, the weight estimation task successfully done by the FastViT model and gained the significant performance.



## 4 Experimental Results

This chapter discussed about the experimental results of the proposed human weight estimation framework. In previous chapter we have discussed the models, its architectures and other details incorporated in the proposed work. Now, in this chapter, we are going to evaluate the proposed framework and analyse the performance of the system. This chapter includes an overview of the hardware and software configurations used to develop the proposed work and execute them successfully. The Celeb-FBI dataset details will be discussed, and it helps to get to know the structure as well as the distribution of the data. It also talks about the relevance of the chosen dataset for this proposed system. The results need to be analysed and to be discussed to know the worthiness of the proposed system. The feature extraction model images and labels if needed will be given in this chapter to understand the output of each model. The details will be given clearly under this chapter to understand the working of the proposed system. Finally, the FastViT model weight estimation will be evaluated using the standard regression performance metrics like Mean Absolute Error (MAE), Root Mean Square Error (RMSE),  $R^2$  Score, and Mean Absolute Percentage Error (MAPE). The graphical illustrations will help to analyse the FastViT model performance further and gives the idea about the overall performance of the proposed system.

### 4.1 System Configuration

The proposed work consists of various computer vision techniques such as instance segmentation, object detection, pose estimation and the hybrid vision transformer model utilized for regression tasks. The appropriate hardware resources are very important for the successful execution of our proposed work. Our work includes not only the dataset which consists of images with varying dimensions with RGB channels, but also the thousands of images feeds into this system. Hence, it is very necessary to check the hardware configurations and use which can optimize the execution of each utilized models in this work. The main advantage of our proposed work is that it builds based on the next-generation vision architecture models. The models used in this work not necessarily needs to get trained from the scratch as it is a pre-trained model. It usually needs normal computing resources, is simply enough for this work. Most of the models doesn't take more amount of time to execute than the common deep learning models.

The Google Colab a free open-source cloud environment is used for our proposed work. The Google Drive utilized for the persistent storage across the systems. In Google Colab, T4 Graphical Processing Unit (GPU) is employed for our proposed work. The system configurations are summarized in the Table 2.

Table 2: System Configurations

| Component        | Specification             |
|------------------|---------------------------|
| Processor        | Google Colab              |
| GPU              | NVIDIA Tesla T4           |
| RAM              | 12-16 GB                  |
| Storage          | Google Drive              |
| Operating System | Linux (Colab Environment) |

The YOLO26-seg model is an instance segmentation model which pre-trained on COCO and ImageNet dataset, and also it shows the speed of 52.8–53.8 ms in CPU then in T4 tensorRT10 shows the speed of 2.1 ms. It gives us the idea of the running time of this model. The RT-DETRV2, an object detection model which is also a pre-trained model on COCO dataset. The estimated runtime for this model in T4 GPU is 108ms whereas for the 2500 images it takes less than 10 mins to detect the objects which is human in our case in the images. The SAM 3 model, a promptable concept segmentation model developed by Meta is notable for its zero-shot performance also it is a pre-trained model to Segment Anything with Concepts (SA-Co) benchmark dataset. The model size is 3.45 GB and can be expected to take, 2921 ms in T4 GPU. The YOLO26-Pose model is a pose estimation model which is a pre-trained model on COCO and ImageNet dataset. The expected runtime for this model on CPU is 562.4–568.4 ms and on T4 GPU is 12–12.4 ms. The FastViT a hybrid vision transformer model using the structural reparameterization which is also a pre-trained model on ImageNet dataset. The expected runtime of this model between 2.387 to 3.583 ms. For the sampled 2500 images from the celeb-fbi dataset, the actual runtime of each of the models are summarized in the Table 3.

Table 3: Runtime of the models

| Models      | Runtime (approx..) |
|-------------|--------------------|
| YOLO26-seg  | < 10 mins          |
| RT-DETRV2   | < 10 mins          |
| SAM 3       | 1.5 hrs            |
| YOLO26-pose | <10 mins           |
| FastViT     | 45 mins            |

Additionally, the Automatic Precision Scaler (AMP) technique is used in this proposed work. It does not simply configure the system; it actively controls the model training by managing the mathematical operations to make it faster and memory-efficient. By default, deep learning models use FP32, whereas the mixed precision technique uses only FP16 for most of the mathematical operations. The challenge here is that some numbers become very small approximately zeros, this is known as the underflow problem. Thus, we have used Gradient Scaling method to avoid such cases. The gradient scaling first scales up the value by multiplying the loss by a large factor then the model runs the core convolutional matrix operations in fast FP16 and at last it scales down the gradients value back to their original size. In Google Colab T4 GPU environment it increases the speed, often up to 2X to 3X in training time. It is memory efficient since it utilizes 50% less memory so that we can easily train our sampled 2500 images. It preserves the accuracy of 32-bit math while doing them in the speed of 16-bit math itself.

#### 4.2 Software Configuration

The proposed work is implemented using various libraries, frameworks, and tools which are open-source platforms available for free to use. The Python programming language used throughout this proposed work. The tools, libraries, and the frameworks are all Python-based tools. The Google Colab is the open-source cloud development environment where we have implemented our whole proposed work. Since, we were used the Google Colab free version T4 GPU, extra care given to the usage of the computational resources. But we have successfully implemented the weight estimation system with these limited computational resources offered by Google Colab T4 GPU because the usage of the next-generational vision architectures models speed of execution and its efficient memory management method.

Several libraries and frameworks have used in this to do the various computer vision tasks like instance segmentation, pose estimation, object detection and the regression tasks. The YOLO26-seg model has been implemented using the Ultralytics framework and the YOLO library. The YOLO26-seg nano variant is utilized in this work. It supports various tasks, but we have utilized for the instance segmentation tasks. The RT-DETRV2, a transformer-based model, has been implemented using the Ultralytics framework and the RT-DETR library. The RT-DETRV2 large variant was utilized in this system. It can be used in the scenarios where the applications require both the high accuracy and real-time performance. The task performed by this model in our proposed system was object detection. The SAM 3 model a promptable concept segmentation model has been implemented with the help of Ultralytics framework, HuggingFace framework and the SAM library. The HuggingFace is the platform where we have requested to access this model for our proposed weight estimation system. The request was accepted by the Meta shortly after half a day. The task performed by this model is instance segmentation using the RT-DETRV2 images as a prompt. The YOLO26-pose a pose estimation model has been implemented in this proposed weight estimation system using the Ultralytics framework and the YOLO library. The YOLO26-pose nano variant was utilized in this work. The task performed by this model is pose estimation of the person in the given image using the keypoints.

The PyTorch used as the core deep learning framework for training the model and increasing the runtime of the model. It can help us to utilize the GPU environment to further accelerate the training. It generates dynamic graphs which helps to know the internal functioning of the model training. Furthermore, it built based on the data structure named tensors which is very similar to the NumPy arrays, but the key difference is the GPU acceleration capabilities. It also supports the automatic differentiation, since it is helpful for deep learning tasks like gradient computation. The timm library is a PyTorch image models which consists of various pre-trained models like ResNet, EfficientNet and the Vision transformers. It is pre-trained on ImageNet dataset. It also consists of collection of utilities, optimizers, data loaders, data augmentation and so on. The data loader has been used to feed the image data to the weight estimation system batch-wise instead of all at a time. The FastViT sa24 variant was

utilized in this system. The data augmentation techniques like horizontal flip, resizing, brightness and contrast adjustments and finally normalization have been applied. The MSE loss function, AdamW optimizer and reduce learning rate on plateau are all have been applied in this system. These are all the utilities we have used in our proposed system using the timm library. All the software configurations used in this proposed system are summarized in the Table 4.

Table 4: Core Software Components

| Component/Library      | Source                     | Version Used               |
|------------------------|----------------------------|----------------------------|
| PyTorch                | PyTorch Official           | 2.1.0                      |
| Torchvision            | PyTorch Official           | 0.16.0                     |
| Timm (FastViT Support) | HuggingFace                | 0.9.12                     |
| RT-DETRV2              | Ultralytics                | rtdestrv2-l (2024 release) |
| SAM 3                  | Meta AI (Hugging Face)     | sam3.pt (2025 release)     |
| YOLO26-seg             | Ultralytics                | yolo26n-seg.pt             |
| YOLO26-Pose            | Ultralytics                | yolo26n-pose.pt            |
| OpenCV                 | OpenCV.org                 | 4.8.0                      |
| HuggingFace Hub        | HuggingFace                | 0.20.3                     |
| TorchInfo              | PyPI                       | 1.8.0                      |
| Joblib                 | PyPI                       | 1.3.2                      |
| Python                 | Python Software Foundation | 3.10                       |
| Ultralytics            | Ultralytics GitHub         | 8.2.0                      |
| Scikit-learn           | Scikit-learn               | 1.3.2                      |

### 4.3 Dataset Description

The Celeb-FBI dataset is the free open-source dataset which is available both on the Kaggle and the HuggingFace. We used the Celeb-FBI dataset taken from the Kaggle. The dataset plays a vital role in building the optimized weight estimation system. Thus, we have used Celeb-FBI dataset carefully designed and applied SMOTE techniques to get the balanced data in the dataset. The images in the dataset are having diverse situations like crowd, background noise presence, lighting variations, etc. The dataset contains 7211 images and each of the files have the metadata which is features like height, age, weight, and gender. The features of this dataset are summarized in the Table 5.

Table 5: Dataset features summary

| Features | Samples | Value                              |
|----------|---------|------------------------------------|
| Height   | 6710    | 4 feet 8 inches to 6 feet 5 inches |
| Weight   | 5941    | 41-110 kg                          |
| Age      | 7139    | 21-80 years                        |
| Gender   | 7211    | Male / Female                      |

Originally, 7222 images found in the dataset then we sampled them into 2500 images for the computational constraints' purposes. The sampled dataset further used in the proposed work. The sampled dataset is split into three components: train (1993 images), val (249 images) and test (250 images). The train and val are used in this proposed method for model building and test set is used for model evaluation. The file naming convention is designed in such a way that it contains the necessary details about the images in the dataset each separated by the underscore. The sampled dataset filename has the metadata needed for model building and the model evaluation. The file naming convention of the image in the dataset is summarized in the Table 6.

Table 6: File naming convention of the images in dataset

| Component | Description              | Example |
|-----------|--------------------------|---------|
| Serial No | Unique ID for each image | 1021    |
| Height    | Feet and inches (h)      | 5.5h    |
| Weight    | Kilograms (w)            | 51w     |
| Age       | Years (a)                | 56a     |
| Gender    | Male/Female              | Male    |

We have created three CSV files named train labels, val labels and test labels where each contains the information about the filename, height, weight, age, and gender. This CSV files will be very helpful in our further steps. The features extracted from the models can be further added to this CSV files separately. The sample images from each split were given in the Figure 2.

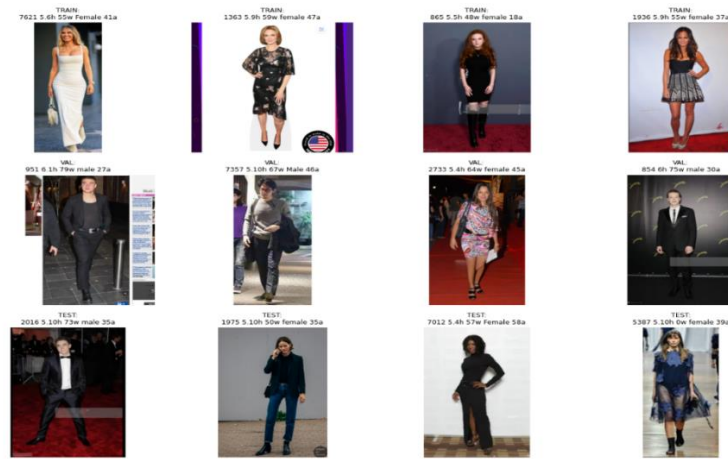


Figure 2: Four sample images from each dataset split

#### 4.4 Results

Initially, we have utilized YOLO26-seg model to do instance segmentation task for the sampled images. The YOLO26-seg model processed the sampled images given to that and able to produce the segmentation using the bounding box with class labels and confident score of the class. However, it wrongly segments a dog as a person in the image, that too with 0.41 confident score. Due to this reason, we have eliminated the YOLO26-seg model from our proposed weight estimation system. The result produced by this model was presented in the Figure 3.



Figure 3: YOLO26-seg model output illustrating incorrect object classification

The YOLO26-seg model failed to capture the context of the image in the sampled dataset. We need a model which captures the global context and as well as able to generate high precision detection in real-time performance. Thus, we have employed RT-DETRV2, a transformer-based model able to capture the global context in the sampled images given to it with high precision result in real-time. The dog problem is now solved by this model and detects the objects correctly with the right confident score as well. The rightly detected image by this model was presented in the Figure 4.



Figure 4: Human Detection using RT-DETRV2 with Refined Bounding Boxes

The RT-DETRV2 model generated images were stored in the Google Drive for further usage in this work. The problem with this result is we are going to estimate the weight for single person, so the background noise like crowd, pictures, flex, posters and any other noises found in the resultant images have to be removed for further process. The model which needs to take the image generated by RT-DETRV2 model as input and gives the cleaned image with background noise removed and consists of single person alone. The SAM 3 model is a promptable concept segmentation model which can take these resultant images as an input and gives the cleaned image. This is the final-cleaned image generated by this model and used in the FastViT model. The cleaned image generated by this model was presented in the Figure 5.



Figure 5: Background Removal using SAM 3 Segmentation

The SAM 3 cleaned images now help us to get the body area by adding all the pixels which are greater than zero. But we need other useful features as well to further enhance the performance of our proposed weight estimation system. The main challenge in estimating the weight is zoom-in problem where a lean person seemed to be bulk due to the zooming. To overcome these issues, we have employed the pose estimation model named YOLO26-pose. The output of the YOLO26-pose model is given in the Figure 6.



Figure 6: Pose Estimation Output with 17 Key points using YOLO26-Pose

The YOLO26-pose model helps us to get both the image with keypoints and labels with 17 keypoints labels, and it is used to extract the key features like height, torso length, shoulder width and the hip width. The 17 keypoints along with its values in the labels folder for a sample image is given in the Table 7.

Table 7: YOLO26-Pose keypoints for a sample image

| Keypoint Index | Body Part      | X (normalized) | Y(normalized) | Confidence (c) |
|----------------|----------------|----------------|---------------|----------------|
| 0              | Nose           | 0.4674         | 0.4904        | 0.4720         |
| 1              | Left Eye       | 0.9708         | 0.4842        | 0.1010         |
| 2              | Right Eye      | 0.9986         | 0.5097        | 0.0819         |
| 3              | Left Ear       | 0.9992         | 0.4618        | 0.0848         |
| 4              | Right Ear      | 0.9951         | 0.5591        | 0.0884         |
| 5              | Left Shoulder  | 0.9913         | 0.4420        | 0.0956         |
| 6              | Right Shoulder | 0.6755         | 0.6263        | 0.2082         |
| 7              | Left Elbow     | 0.9991         | 0.4061        | 0.2151         |
| 8              | Right Elbow    | 0.9991         | 0.6576        | 0.3658         |
| 9              | Left Wrist     | 0.9970         | 0.3779        | 0.3702         |
| 10             | Right Wrist    | 0.9922         | 0.6855        | 0.5128         |
| 11             | Left Hip       | 0.9965         | 0.3459        | 0.4965         |
| 12             | Right Hip      | 0.9927         | 0.5822        | 0.4462         |
| 13             | Left Knee      | 0.9992         | 0.4344        | 0.4448         |
| 14             | Right Knee     | 0.9993         | 0.5818        | 0.6669         |
| 15             | Left Ankle     | 0.9986         | 0.3696        | 0.6737         |
| 16             | Right Ankle    | 0.9983         | 0.5400        | 0.8475         |

The proposed framework for the human body weight estimation, the multi-stage feature extraction pipeline was successfully executed on the sampled dataset consists of 2500 images out of them 2492 images were selected after preprocessing. The initial detection stage using the RT-DETRV2 model successfully processed all the images which were given to them. The segmentation model SAM 3 model also processed all the images without any failure. The cleaned SAM 3 images were given to the YOLO26-Pose model, which also processed all the images without any failure. However, pose labels were obtained for only 2487 images with 5 images lacks the corresponding label outputs. These may happen due to the conditions like low visibility, incorrect human posture, etc. To maintain the integrity of the dataset and the loss of this image data, default feature values were assigned for such cases at the time of feature extraction. Overall, the proposed multi-stage feature extraction pipeline shows certain level of reliability successfully processing all images while maintaining the success rate of 99.8% for the YOLO26-pose model along with the average pose confident score of 0.41.

The final stage of our proposed weight estimation pipeline is the trained FastViT model and its evaluation. The FastViT model here, in this case we have utilized this for regression tasks. The standard regression model has some common performance metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Coefficient of determination ( $R^2$  score), and Mean Absolute Percentage Error (MAPE). These are all the metrics used and analysed to recognize the strength and limitation of our proposed weight estimation system.

The MAE has been computed by taking the average absolute difference between the predicted value of our model and the actual value found in the CSV files. It helps us to get the direct and interpretable measure of the error in the same unit as the target variable weight in our case. The MAE value we got from our trained model was 6.37 approximately 6. It indicates that if the new image will be given to the model, it estimates the weight of the person in  $\pm 6$  kg. For example, if the person actual weight is 76 kg, then the model weight estimation will be around 70-82 kg. In my analysis, we have found out the model performance was reasonably well since it has decent MAE value for the Celeb-FBI dataset. The formula for MAE is given in the Eq. (1).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

The RMSE is a sensitive to outlier metric which penalizes the large errors. It is computed by taking average squared differences between the predicted value of our model and the actual value found in the CSV files. It doesn't treat all the errors equally. It conveys how our model performed for certain complex cases. The RMSE of our trained model was found to be 8.36 which indicates that for some of the cases the weight estimation of our model extends to approximately  $\pm 8$  kg. However, the difference between the MAE and RMSE is low and RMSE value is less than 10 so it is acceptable for our model. The formula for RMSE is given in the Eq. (2).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

The  $R^2$  score is a metric which tries to explain how well the model able to explain the variance in the actual values. In simple words, it is a metric which tells us about how well the model interpret the logic of the given problem. For our proposed weight estimation FastViT model we got  $R^2$  score of 0.69 value which means 69% nearly 70% the model able to understand the weight estimation logic remaining 30% is the error or noise. Our FastViT model gives the moderate level performance, which can be useful for our proposed weight estimation work. The formula for  $R^2$  score is given in the Eq. (3).

$$R^2 \text{ score} = 1 - \left( \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \right) \quad (3)$$

The MAPE is a kind of relative error expressed in terms of percentage. It tells about the relative percentage error, which is how much it deviates from the actual value. If the MAPE value is less than 10% means it comes under the highly accurate model. Our model got MAPE of 9.61% which tells it deviates this much from the actual value, and it is less than 10% which means our model comes under the highly accurate category. The formula for MAPE is given in the Eq. (4).

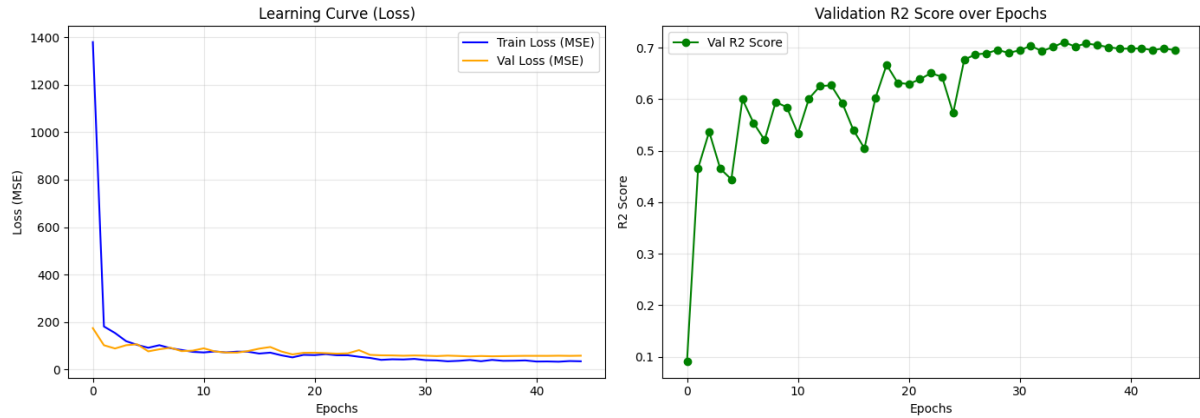
$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (4)$$

The performance metrics obtained for the FastViT sa\_24 model are summarized in Table 8, providing a correct evaluation of the model's prediction capability.

Table 8: Performance Metrics

| Performance Metrics | Value |
|---------------------|-------|
| MAE                 | 6.37  |
| RMSE                | 8.36  |
| $R^2$ Score         | 0.69  |
| MAPE                | 9.61% |

To better understand the model training behaviour, we need to further study about the performance of the model. The model training needs to be studied dynamically to show how our model performs and how relevant it is. The figure 7 consists of two components: Learning Curve and Validation  $R^2$  score over Epochs. These two components helps us to understand the impact of all the training configurations especially the learning rate, optimizer, loss function and the early stopping. When it comes to training the model whether it is regression, classification or any other tasks we need to first look for overfitting or underfitting. The overfitting is the challenge where the model learns the training data given to it very well gives low error for the training data whereas it provides high error on the new unseen data. The underfitting happens when a model is too simple to learn the underlying patterns of the data, it performs poorly on both the training data and test data. These two problems are very common and needs to be analysed before further processing. The Figure 7 contains the two components are given below.

Figure 7: Learning curve and validation  $R^2$  score over training epochs

The first component of the Figure 7 shows the Learning Curve (Train loss vs Validation loss). The MSE loss function used in this graph. Initially in the epoch ranges from 0-2 which shows high loss both in the training and validation. The model weights here are randomly initialized, which clearly shows the no understanding of the body features yet. The model doesn't start its training so far in this phase. The Rapid drop phase in the epoch ranges from 1-10, which shows clarifies that there is a huge drop in both training and validation loss. The model started to learn faster in the first 10 epochs itself, which means our choice of optimizer works reasonably well for our model. The AdamW optimizer is a popular stochastic gradient descent optimizer, it decouples the weight decay from the update of gradient, leading to better regularization and generalization. It applies the weight decay straightly to the weights, rather than loss functions and the gradient calculations. It is particularly effective for training complex models like Transformer models. Likewise, it started to update the weights in this phase.

The stable learning phase starts in the epoch ranges from 10-30, where the loss gradually decreases and shows small fluctuations. The learning rate scheduler we have used here is ReduceLROnPlateau which reduces the learning rate slowly when there it attains the state of plateau. When the validation loss stops improving, the learning rate gradually reduced. This is the phase where the model stops making the large updates and started fine-tuning the weights carefully. The smooth convergence phase started in the ranges from 30 to 45 epochs where the training loss and the validation loss almost parallel, there is no large gap. This is the place where we can clearly see that this minimal gap between the train and validation loss indicates that the model doesn't suffer from the overtraining as well as the underfitting. The model training sets to 50 epochs, but it stops at 45th epoch because of the early stopping triggered. The early stopping trigger set for 10 epochs which means after no improvements for the 10 epochs, and it stops at 45th epoch which means at 35th epoch the model stops learning further, and it achieves the plateau.



The second component of the Figure 7 is Validation  $R^2$  score over Epochs graph. After the analysis made from the graph, there are visible phases where model behaves in a certain way. In the initial phase, the value is low approximately 0.1 this indicates that the predictions made by the FastViT model looks like almost random there is no clear sign of understanding of the extracted features. The rapid growth phase started from the 1st epoch to 10th epoch, the value jumps to 0.6. This is the clear sign that the model learning ability and the understanding of the weight estimation logic skyrocketed in the initial phase itself. It demonstrates that the extensive feature extraction pipeline used in the proposed weight estimation framework works reasonably well. The feature engineering impact clearly visible in this phase. The fluctuation phase occurs in the range of 10–25 epochs, where the value oscillates between 0.5 and 0.65. Due to the data augmentation techniques applied in this model it started seeing the different variations in the image, the model oscillates back and forth in this phase. The stabilization phase finally seen in the 25th epoch to 45th epoch where the value started stabilizes around 0.69-0.70. It shows the model architecture advantage the hybrid vision transformer model where the CNN part captures the local features and the transformer part captures the global structure. Overall, from this Figure 7 clearly shows that there is no overfitting and underfitting signal and conveys the generalization of the model looks reasonably well. The figure 8 will clearly demonstrate us to know the model performance in detail. It will help us to know the model performance and its lag where it can fit as well as where it struggles to fit. The Figure 8 actual vs predicted weights plot is given below.

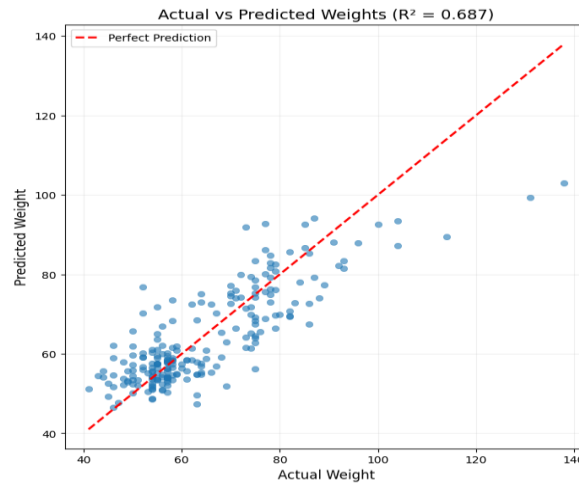


Figure 8: Actual vs predicted weight values

The graph contains three things: X-axis, Y-axis and a red dashed line. The X-axis represents the actual weight of the person in the image, Y-axis represents the predicted weight by the FastViT model of the person in the image, the red dashed line represents a perfect prediction line and every dot points in the graph represents the person in the sampled dataset. The core idea of this plot is that closer the points are to the diagonal line, the model prediction accuracy will be better. The overall pattern in the graph was that all the points are clustered around the diagonal line, not perfectly on the line, but it seems to be close enough. It means the model had learned the relationship in the weight estimation logic but still it suffers from some prediction error which is very common in the regression tasks. The value 0.69 for the  $R^2$  score clearly explains that the trained model can explain 69% of the variance in weight of the person. The remaining 31% would be noise, the missing features and the complexity. With this percentage value, the model demonstrates a strong positive correlation between the actual and the predicted values. It seems to be effectively learning the underlying patterns and its relationships.

The error distribution pattern is very important for us to know the place where the model needs to be improved. When analysed carefully we have found out that in the lower weight range around 50-65 kg the points were tightly clustered and seems to be very close to the diagonal line. From this, we can infer that the trained model can give better performance for estimating the weight of the lean individuals very accurately. In the mid-range around 65-90 kg there was some slight spread in the graph, but it is still near the diagonal line. It shows us that it gives good performance for the mid-range individuals with some slightly minor deviation. In the high weight range, which is more than 90+ kg, the points in the graph were more likely to deviate from the diagonal line. The underprediction

for this individual's range are clearly visible. It demonstrates that the model struggles to estimate the weight of the person with 90+ kg. Due to the imbalance in the celeb-fbi dataset, this happens for the heavier individuals. Maybe, more samples in the normal weight range in the celeb-fbi dataset and very less data for the extreme weights. Hence, the model becomes biased toward the common weight range. The additional features will be required for us to overcome these issues by having the body fat distribution, muscle density and the depth information. The underprediction trend clearly visible in the graph where in the top-right region the actual weight is high but predicted weight slightly lower, and it is known as regression toward the mean. The model avoids the extreme conditions, and it will play safe by predicting the weights closer to the average weight. Overall, from this Figure 8 it clearly shows us that our model performs reliably with minor limitation.

The plot in the Figure 9 shows the residual plot (Predicted VS Actual) the X-axis represents the predicted weight by the model, the Y-axis represents the residuals which had been derived by differencing the actual weight and the predicted weight, the red line represents the perfect prediction and each point in them represents one prediction error. The core idea behind this plot is that used to check whether the trained model errors are random or showing any patterns. The residual plot is given in the Figure 9 below.

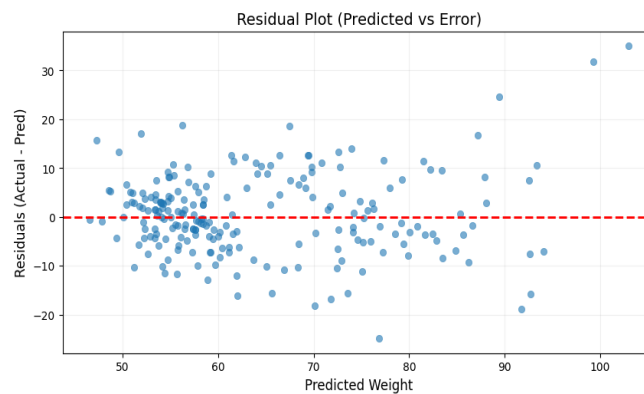


Figure 9: Residual plot showing prediction errors

The overall distribution of the residuals in the plots was scattered all around the zero line. In this plot, it seems to be both the positive and the negative errors were present. It clearly demonstrates not only that the model is not biased in any one particular direction, but also the predictions made by the FastViT model were balanced. From this residual plot, we also observed that there is no clear curve, no clear sign of patterns, and it is random, which is actually good for our model. It shows us that the model has captured the most of the relationships and the remaining errors were all just the random noise. There are two cases found to be present in this plot, which is positive residuals and the negative residuals. The positive residuals tell us about that the predicted weight of the person by our model is actually less than the actual weight of the person. It means that our model underestimated the weight of the person. The negative residuals tell us about that the predicted weight of the person by our model is actually greater than the actual weight of the person. It means that our model overestimated the weight of the person. After analysing the spread of the errors in the plot it seems to be for low predicted values around 50-60 kg errors are small and tightly clustered around the zero. It means that it is estimating the weight of the lean persons with high accuracy. The mid-range around 60-80 kg shows the moderate spread which means there is a deviation found to be there, still it is acceptable. However, for high predicted values around 80-100+ kg shows there is a large spread and errors are even up to -25 or +35. It clearly demonstrates that the model becomes less stable for the heavier individuals. Overall, from this Figure 9 we got the key insights about the performance of the model even if it has some limitation still it can really work well in real-world.

The Figure 10 the last figure which shows the distribution of the prediction errors with histogram and Kernel Density Estimation (KDE). The graph contains the X-axis represents the prediction error computed by taking difference between the actual weight and the predicted weight, the Y-axis represents the frequency, the bars represent that how often a certain occurs and the smooth curve KDE represents the overall distribution shape of the prediction errors. The core idea behind this plot is that how the prediction errors are distributed to check

whether our FastViT model is biased or well-balanced one. The distribution of the prediction errors shown in the Figure 10 below.

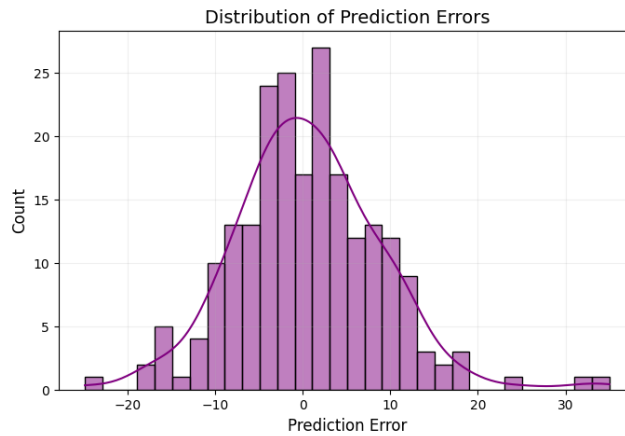


Figure 10: Distribution of prediction errors

The shape of the distribution observed to be a bell-shaped curve centred around zero, which means it is approximately a normal or Gaussian distribution. It shows us that the most of the errors made by the model found to be small, the large errors are found to be rare, and it infers that the model is statistically stable one not biased one. The centring around zero that is the peak is near zero error. It simply conveys that the predicted weight value of the person approximately closer to the actual weight of the person. After this analysis, we can understand that our model is not always overestimating the weight of the person, at the same time not always underestimating the weight of the person. There is no systematic bias found in this plot. The spread of the errors are needs to be observed very carefully, which is very important to get the key insights from this plot. The majority of the errors found to be between -10 to 10 kg, and most predictions are considered to be reasonably accurate. There are few extreme errors around -25 to 35 kg, which means the presence of the outliers in our proposed work. The plot seems to show slight skewness when closely looked it found to be a right tail slightly longer which indicates that the model could have a slight tendency to underestimate the weight of the person in the image. The smooth KDE curve observed and shows that there are no irregular spikes and any traces of multi-peak behaviour which means that model errors are all consistent there is no hidden bias clusters at last the model learns stable.

Overall, the rigorous analysis of our model shows that our proposed weight estimation framework found to be performs reasonably well, and it can be utilized for the real-world implementation. The extensive feature extraction work along with the use of hybrid vision transformer model performs well for the weight estimation tasks.

## 5 Conclusion and Future Work

The proposed framework for estimating the human body weight exhibits the strong performance by effectively utilizing and integrating the next-generation vision architectures with an extensive feature extraction pipeline. The use of advanced models for each computer vision tasks like segmentation, object detection, pose estimation and also for the regression tasks enables us to extract the meaningful features like body area, height, shoulder width, hip width and torso length. The Celeb-FBI dataset has been successfully explored and utilized well to illustrate that the image-based weight estimation possibility. The FastViT a hybrid vision transformer model has been analysed using the various regression performance metrics like MAE, RMSE,  $R^2$  Score and MAPE. The overall results analysed by us shows that the proposed weight estimation framework for the human body is capable of capturing the relationship between the human body structure and the weight appropriately. It makes our proposed work to be a viable approach for real-world applications.

Even though our proposed framework for the human body weight estimation works well, but it shows us the place where it does not work really well. Our model performs well for the lean and moderate weight individuals, but there are high errors and not correctly estimating the weight of the human body for the overweight individuals with over 90+ kg weight. This issue is analysed by us, and it has been observed that several data in the dataset found to be inconsistent to the ground truth. For example, a tall lean person, possibly his weight ranges from 70-90 kg labelled as 120 kg. It helps us to shift our focus more on the data preprocessing stage. The more advanced data preprocessing needs to be employed like outlier detection to further improve the performance of the model. In addition to that, further improvements can also be achieved by the use of additional meaningful features that are possibly ale to extract from our segmentation, object detection and pose estimation models. By making the improvements, we have observed and discussed so far not only improve the human body weight estimation accuracy but also enhance the model's capability to generalize well to the real-world conditions.

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